

Hybrid CP-SAT and Quantum-Inspired Optimization for Airline Disruption Recovery with Explainable Decision Support

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Abstract

Airline disruption management requires solving tightly coupled combinatorial optimization problems—aircraft assignment, crew scheduling, and delay propagation—under strict regulatory constraints and real-time operational pressure. This paper presents a hybrid decision support framework that integrates Google OR-Tools CP-SAT constraint programming with a Quantum-Inspired Genetic Algorithm (QIGA) through a warm-start mechanism, augmented by a SHAP-based explainability layer for transparent decision-making. The CP-SAT solver generates feasible solutions that strictly enforce EASA Flight Time Limitation (FTL) regulations and operational constraints including aircraft continuity, crew continuity, gate capacity, and minimum connection times. These solutions are subsequently refined by QIGA, which leverages quantum-rotation-gate-driven population evolution to escape local optima within limited computational budgets. An explainability module combining TreeSHAP feature attribution with constraint-driven decision reason codes provides interpretable rationale for each operational decision, addressing EASA AMC 20-42 transparency requirements. Experimental evaluation on a

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150-flight hub-and-spoke network demonstrates that the hybrid approach achieves a 5% improvement in operational profit over standalone CP-SAT ($p = 0.007$), reduces flight cancellations by 20%, and maintains zero FTL violations across all test scenarios. The system converges to solutions within 2.1% of optimality under a 60-second time limit, confirming practical viability for real-time airline operations control.

Keywords: Airline Disruption Management, Constraint Programming, CP-SAT, Quantum-Inspired Genetic Algorithm, Explainable AI, SHAP, EASA Flight Time Limitations, Irregular Operations

1 1. Introduction

2 Commercial aviation transported approximately 4.5 billion passengers and ex-
3 ecuted 8.8 million scheduled flights in 2023 [17]. This operational scale gen-
4 erates tightly coupled decision spaces encompassing flight scheduling, aircraft-
5 crew assignments, airport slot coordination, maintenance windows, and passen-
6 ger connections—each individually NP-hard in combinatorial complexity [3, 5].
7 Historical NEXTOR estimates put the annual cost of US flight delays at approxi-
8 mately \$32.9 billion (2007 USD), a figure that remains indicative of the systemic
9 scale even after inflation adjustment [9], with significant portions attributable to
10 suboptimal resource allocation and delayed operational decisions rather than ex-
11 ternal factors [4]. AhmadBeygi et al. [2] demonstrated that crew duty chain con-
12 tinuity plays a critical role in delay propagation across airline networks.

13 Traditional approaches to disruption management rely on either exact opti-
14 mization methods—such as Mixed-Integer Linear Programming (MILP) and con-

15 straint programming—or metaheuristic techniques including genetic algorithms
16 and local search. While exact methods provide strong feasibility guarantees and
17 can rigorously enforce operational constraints, they often struggle to scale un-
18 der the tight time limits imposed by real-world airline operations control centers
19 [6, 15]. Conversely, metaheuristic approaches offer improved exploration of large
20 solution spaces but may fail to guarantee feasibility with respect to complex reg-
21 ulatory constraints or converge prematurely to suboptimal solutions [19, 31].

22 Recent advances in constraint programming, particularly Google OR-Tools
23 CP-SAT, which combines constraint propagation with Boolean satisfiability (SAT)-
24 based conflict-driven clause learning, have demonstrated significant potential in
25 solving large-scale combinatorial scheduling problems [25, 18]. In parallel, quantum-
26 inspired evolutionary algorithms have emerged as promising metaheuristics that
27 enhance population diversity through probabilistic representations inspired by quan-
28 tum superposition [13, 34].

29 Another critical limitation of existing decision support systems lies in their
30 lack of transparency. The European Union Aviation Safety Agency (EASA) AMC
31 20-42 guideline mandates that AI/ML-based decision systems in aviation pro-
32 vide explainability, traceability, and robustness [11]. SHAP [21], LIME [28], and
33 counterfactual explanations [32] represent post-hoc explainability methods that
34 address these requirements.

35 Individual components of this pipeline (CP-SAT scheduling, quantum-inspired
36 metaheuristics, SHAP attribution) have each appeared in prior work, typically in
37 isolation. The contribution of this paper is not the invention of a new component,

38 but rather a **tightly coupled integration** for which we can offer (i) a formal warm-
 39 start non-degradation guarantee (Proposition 1) that makes the CP-SAT $\rightarrow$ QIGA
 40 hand-off provably safe under a fixed time budget, and (ii) a structured mapping
 41 between optimization constraints and post-hoc explanations aligned with EASA
 42 AMC 20-42 traceability requirements. To our knowledge, no prior study com-
 43 bines all three elements under a single formally-analyzed framework targeting
 44 airline disruption recovery. Unlike classical recovery optimization approaches
 45 that lack explainability [27] or learning-based predictors that cannot enforce hard-
 46 constraint satisfaction [14], the proposed framework ensures **mathematical feasi-**
 47 **bility** of all solutions while providing **auditable decision rationale** aligned with
 48 emerging aviation AI regulation (EASA AMC 20-42). Furthermore, whereas
 49 existing quantum-inspired optimization studies [34] evaluate QIGA in isolation,
 50 we demonstrate that its effectiveness is **contingent on warm-start initialization**
 51 from an exact solver—a finding with implications for hybrid algorithm design
 52 beyond the aviation domain.

53 The main contributions are:

- 54 1. **Hybrid Optimization Framework:** A novel two-stage integration of CP-
 55 SAT and QIGA for airline disruption recovery. We formally prove (Proposi-
 56 tion 1) that warm-start initialization guarantees monotonic non-degradation
 57 of solution quality, combining exact feasibility with enhanced exploratory
 58 search.
- 59 2. **EASA-Compliant Constraint Formulation:** Direct encoding of EASA
 60 CAT.OP.MPA.210 Flight Time Limitations as hard constraints in the CP-

61 SAT model, with automatic generation of decision reason codes via post-
62 solve constraint slack analysis.

63 **3. Integrated Explainable Decision Support:** A dual-layer explainability ar-
64 chitecture combining TreeSHAP feature attribution (predictive transparency)
65 with structured constraint-based decision reasoning (optimization transparency),
66 providing auditable rationale aligned with EASA AMC 20-42.

67 **4. Comprehensive Experimental Evaluation:** Rigorous performance analy-
68 sis with statistical significance testing (Wilcoxon signed-rank, $p = 0.007$),
69 ablation studies, and multi-scenario robustness evaluation demonstrating
70 consistent improvements under diverse disruption conditions.

71 The remainder of this paper is organized as follows. Section 2 reviews related
72 work. Section 3 presents the mathematical formulation. Section 4 describes the
73 hybrid optimization framework. Section 5 details the explainability layer. Sec-
74 tion 6 provides experimental results. Section 7 discusses findings and limitations.
75 Section 8 concludes the paper.

76 **2. Related Work**

77 *2.1. Exact Optimization Methods*

78 The classical formulation of crew pairing as a Set Partitioning Problem (SPP)
79 with column generation remains the gold standard for offline planning [8, 6]. Fleet
80 assignment is typically modeled as MILP [15, 1, 30]. While these methods guar-
81 antee global optimality, solution times range from hours to days for large-scale
82 instances, rendering them impractical for real-time disruption recovery. Yen and

83 Birge [33] extended this framework to stochastic programming, though computa-
84 tional costs escalate with scenario count.

85 2.2. *Constraint Programming*

86 Constraint programming offers natural modeling for multi-constrained schedul-
87 ing [29]. Google OR-Tools CP-SAT combines lazy clause generation, conflict-
88 driven clause learning, and portfolio search, achieving top performance in multi-
89 ple MiniZinc Challenge benchmarks [25]. Laborie et al. [18] demonstrated that
90 CP Optimizer’s interval variable abstractions enable more compact modeling for
91 scheduling problems compared to MILP.

92 2.3. *Metaheuristic and Quantum-Inspired Methods*

93 Genetic algorithms for crew scheduling were pioneered by Levine [19] and
94 extended by Ozdemir and Mohan [24]. The Quantum-Inspired Evolutionary Al-
95 gorithm (QIEA) proposed by Han and Kim [13] encodes chromosomes as Q-bits,
96 maintaining population diversity longer than classical GA. Zhang [34] reviewed
97 quantum-inspired evolutionary algorithms across combinatorial domains and re-
98 ported 15–30% faster convergence on large instances relative to classical GAs.

99 2.4. *Explainable AI in Aviation*

100 SHAP [21] provides game-theoretic feature attribution; TreeSHAP [22] com-
101 puts exact Shapley values for tree-based models in polynomial time. Counterfac-
102 tual explanations complement feature attribution by identifying minimal action-
103 able changes that flip a prediction [32].

104 2.5. *Hybrid Approaches and Research Gap*

105 Rosenberger et al. [27] formulated aircraft rerouting as a large-scale integer
106 program for recovery, but without explainability or learning-based demand priors.
107 Jacquillat and Odoni [20] integrate scheduling and operations for airport congestion
108 mitigation but do not address airline-side crew or FTL constraints. Barredo
109 Arrieta et al. [12] survey XAI taxonomies relevant to safety-critical AI but without
110 integration into combinatorial optimization pipelines. The simultaneous combination
111 of CP-SAT exact optimization with quantum-inspired metaheuristic refinement and
112 integrated multi-level explainability for airline disruption recovery
113 remains unexplored in the literature—a gap this paper addresses.

114 Specifically, three critical gaps emerge from the literature: (i) the **optimality–**
115 **speed–explainability trilemma**, where exact methods guarantee optimality but
116 lack transparency, while learning-based methods are fast but cannot ensure regulatory
117 compliance; (ii) the **absence of open-source EASA FTL integration** in
118 CP models with automatic constraint attribution; and (iii) the **lack of empirical**
119 **evidence** on whether warm-start coupling between exact and quantum-inspired
120 solvers yields synergistic benefits beyond standalone performance.

121 Table 1 provides a comparative positioning of the proposed approach.

Table 1: Comparative positioning of the proposed approach in the literature.

Approach	Source	Problem	Scale	Time	Optimality	XAI
Column Generation	[6]	Crew pairing	≤ 5000	Hours	Global	—
MILP (Gurobi)	[26]	Fleet assign.	≤ 2000	Minutes	Global	—
CP + CBLS	[18]	Scheduling	Medium	Minutes	Near-opt.	—
GA	[19]	Crew pairing	Medium	Seconds	Heuristic	—
QIEA	[13]	Combinatorial	Variable	Seconds	Heuristic	—
ML prediction	[14]	Delay prop.	Network	Offline	Predictive	—
Rerouting IP	[27]	Recovery	Medium	Minutes	Near-opt.	—
This paper	—	IROPS	≤ 500	≤ 60 s	CP+QIGA	✓

122 3. Problem Formulation

123 3.1. Problem Description

124 Airline disruption recovery is modeled as an integrated optimization problem
125 that jointly considers aircraft assignment, crew scheduling, and delay management
126 under operational and regulatory constraints. The system is represented using a
127 time–space network (TSN), where nodes correspond to airport–time pairs and arcs
128 represent feasible transitions such as flight legs, ground waiting, and maintenance
129 activities.

130 3.2. Sets and Decision Variables

131 Let F , K , and C denote the sets of flights, aircraft, and crew, respectively, with
132 A denoting the set of airports and T the set of time slots (15-minute intervals).

Table 2: Decision variables of the CP-SAT model.

Symbol	Type	Description	Domain
$x_{f,k}$	Binary	Flight f assigned to aircraft k	$f \in F, k \in K$
$y_{f,c}$	Binary	Flight f assigned to crew c	$f \in F, c \in C$
z_f	Binary	Flight f is cancelled	$f \in F$
d_f	Integer	Delay assigned to flight f (min)	$0 \leq d_f \leq D_{\max}$
δ_f	Integer	Actual departure time of f	—
$g_{f,a,t}$	Binary	f uses gate at airport a , time t	$(f, a, t) \in \mathcal{G}$

133 3.3. Constraints

134 The model enforces ten constraint families ensuring operational feasibility and
135 regulatory compliance (Table 3).

Table 3: Constraint families.

Constraint	ID	Description	Type
Assignment uniqueness	C ₁	Each flight $\rightarrow$ at most one aircraft	Hard
Crew uniqueness	C ₂	Each flight $\rightarrow$ at most one crew	Hard
Aircraft capacity	C ₃	Capacity $\geq$ passenger count	Hard
Aircraft continuity	C ₄	Same aircraft connects at same airport	Hard
Crew continuity	C ₅	Same crew connects at same airport	Hard
EASA FTL	C ₆	Crew cumulative duty $\leq$ 600 min	Hard
Gate capacity	C ₇	Max 1 aircraft/gate/time slot	Hard
Slot capacity	C ₈	Hourly movements $\leq$ airport capacity	Hard
Min. Connection Time	C ₉	Passenger connection $\geq$ MCT	Hard
Cancel consistency	C ₁₀	Cancelled $\Rightarrow$ no assignments	Hard

136

(C₁) Assignment uniqueness:

$$\sum_{k \in K} x_{f,k} + z_f = 1, \quad \forall f \in F \quad (1)$$

137

(C₂) Crew uniqueness:

$$\sum_{c \in C} y_{f,c} \leq 1 - z_f, \quad \forall f \in F \quad (2)$$

138 **(C₃) Aircraft capacity:**

$$\sum_{k \in K} \text{cap}_k \cdot x_{f,k} \geq \text{pax}_f \cdot (1 - z_f), \quad \forall f \in F \quad (3)$$

139 **(C₄) Aircraft continuity:** For consecutive flights f_1, f_2 on the same aircraft

140 k :

$$x_{f_1,k} + x_{f_2,k} \leq 1 \quad \text{if } \text{dest}_{f_1} \neq \text{orig}_{f_2} \quad (4)$$

141 **(C₆) EASA FTL (conservative proxy):**

$$\sum_{f \in F} y_{f,c} \cdot \text{block}_f \leq \bar{B}, \quad \forall c \in C, \quad (5)$$

142 where $\bar{B} = 600$ min is used as a *conservative uniform ceiling* derived from the
 143 tightest single-crew-type entry of EASA CAT.OP.MPA.210 under a one-sector,
 144 non-WOCL scenario. This proxy is deliberately restrictive: any schedule satisfy-
 145 ing (5) is also feasible under the full FDP $\times$ sector $\times$ WOCL matrix, so the model
 146 cannot produce FTL violations when the complete table is substituted. We re-
 147 tain (5) in the main experiments because it preserves the combinatorial structure
 148 exercised by CP-SAT while yielding a regulator-safe over-approximation; a richer
 149 table-driven formulation is a drop-in replacement that we discuss in Section 7.

150 **(C₉) Minimum Connection Time:** For passenger connection $(f_{\text{in}}, f_{\text{out}})$:

$$\delta_{f_{\text{out}}} - (\delta_{f_{\text{in}}} + \text{block}_{f_{\text{in}}}) \geq \text{MCT}_a \cdot (1 - z_{f_{\text{in}}} - z_{f_{\text{out}}}) \quad (6)$$

151 3.4. Objective Function

152 The objective minimizes total operational cost:

$$\min \sum_{f \in F} \left[\alpha \cdot z_f \cdot R_f + \beta \cdot d_f \cdot w_f + \gamma \sum_{k \in K} x_{f,k} \cdot \text{fuel}_{f,k} + \mu \cdot z_f \cdot \text{slotPen}_f \right] \quad (7)$$

153 where R_f is the revenue loss from cancellation, w_f is the delay cost per minute
 154 (weighted by passenger volume and connection criticality), $\text{fuel}_{f,k}$ is the fuel cost,
 155 and $\alpha, \beta, \gamma, \mu$ are strategy-dependent weights (Table 4).

Table 4: Strategy weight configurations.

Strategy	α (cancel)	β (delay)	γ (fuel)	μ (slot)
PROFIT	1.0	1.0	0.3	0.5
ECO	0.8	1.2	1.5	0.4
RESILIENCE	1.5	0.8	0.4	1.0

156 3.5. Complexity Analysis

157 The model has $O(|F| \cdot (|K| + |C|))$ variables and $O(|F|^2)$ constraints. For the
 158 experimental instance ($|F| = 150$, $|K| = 20$, $|C| = 40$): $\sim 9,000$ variables and
 159 $\sim 22,500$ constraints. QIGA complexity: $O(G_{\max} \cdot N \cdot |F|)$ with typical $G_{\max} =$
 160 200 , $N = 50$.

161 4. Hybrid Optimization Framework

162 4.1. Overview

163 The proposed framework operates in two stages (Fig. 1): (i) the CP-SAT solver
164 generates a high-quality feasible solution under strict constraint enforcement, and
165 (ii) QIGA refines this solution through population-based evolutionary search ini-
166 tialized via warm-start. When CP-SAT returns `TIMEOUT` or `INFEASIBLE`, the
167 system automatically transitions to QIGA for recovery.

168 4.2. Stage 1: CP-SAT Feasible Solution Generation

169 The first stage employs the CP-SAT solver from Google OR-Tools [25]. CP-
170 SAT combines constraint propagation with SAT-based conflict-driven clause learn-
171 ing, enabling efficient exploration of large combinatorial spaces. The solver op-
172 erates under a configurable time budget (default: 60 seconds) and guarantees that
173 all hard constraints—including aircraft continuity (C_4) and EASA FTL (C_6)—are
174 strictly satisfied. Let S_{CP} denote the best feasible solution obtained within the
175 time limit.

176 4.3. Stage 2: Quantum-Inspired Genetic Algorithm (QIGA)

177 Each individual in the QIGA population is encoded using a vector of Q-bits:

$$q_i = (\alpha_i, \beta_i), \quad |\alpha_i|^2 + |\beta_i|^2 = 1 \quad (8)$$

178 Each chromosome $\xi = (\xi_1, \xi_2, \dots, \xi_n)$ encodes an action vector where $\xi_i \in$
179 $\{\text{KEEP}, \text{DELAY}_{15}, \text{DELAY}_{30}, \text{DELAY}_{60}, \text{CANCEL}, \text{SWAP}\}$.

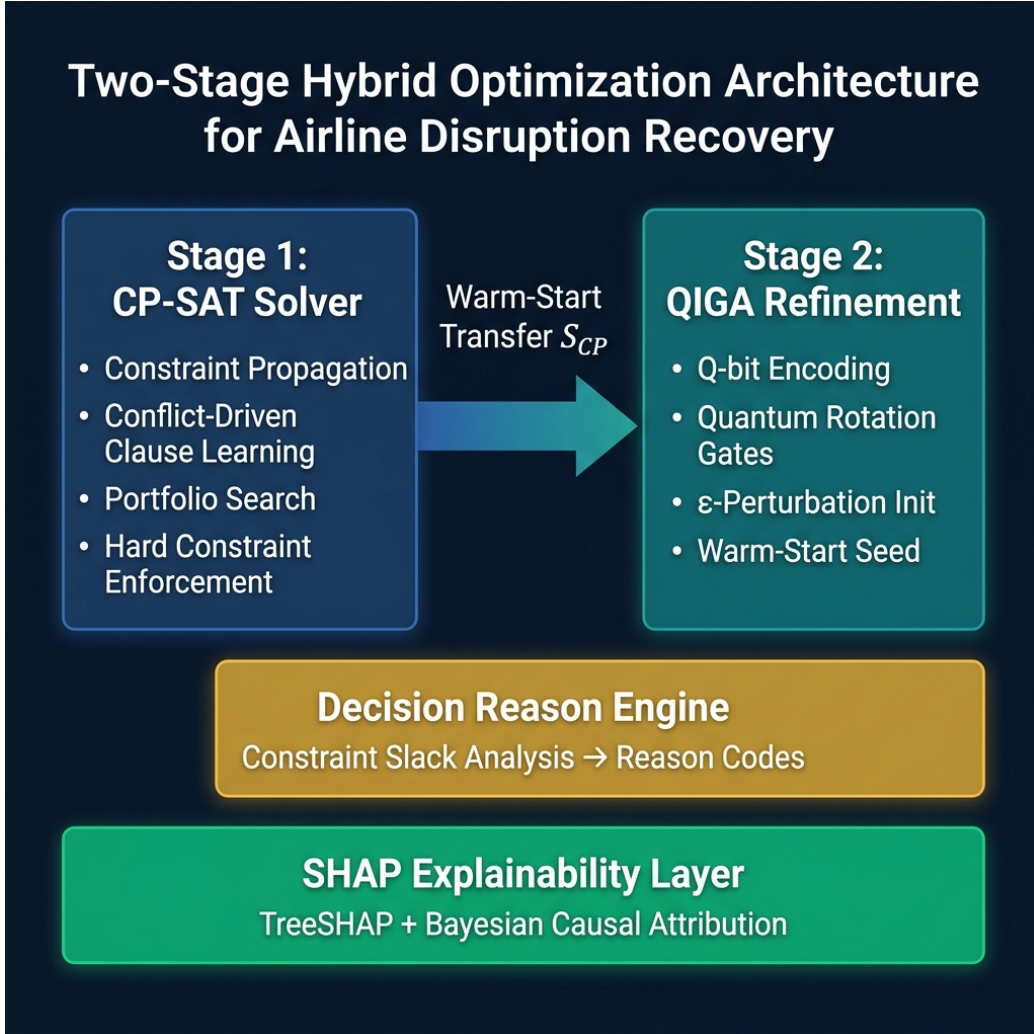


Figure 1: Two-stage hybrid optimization architecture. The CP-SAT solver generates feasible solutions enforcing all hard constraints; QIGA refines via quantum-rotation-gate-driven population evolution initialized with the CP-SAT solution.

180

Population evolution is governed by quantum rotation gates:

$$\begin{pmatrix} \alpha'_i \\ \beta'_i \end{pmatrix} = \begin{pmatrix} \cos \Delta\theta & -\sin \Delta\theta \\ \sin \Delta\theta & \cos \Delta\theta \end{pmatrix} \begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} \quad (9)$$

181 where $\Delta\theta$ is adaptively determined based on the fitness difference between the
 182 current solution and the best-known solution [13].

183 **Fitness function:**

$$\mathcal{F}(\xi) = -\text{Objective}(\xi) - \lambda \cdot \text{Violations}(\xi) \quad (10)$$

184 4.4. Warm-Start Integration

185 The warm-start mechanism tightly couples CP-SAT and QIGA:

- 186 1. S_{CP} is used as the **first individual** in the QIGA population.
- 187 2. The remaining $N - 1$ individuals are generated via ϵ -perturbation around
 188 S_{CP} .
- 189 3. Q-bits are initialized with biased amplitudes $(\alpha, \beta) = (0.95, 0.31)$, concen-
 190 trating the search near the feasible region.

191 This provides **feasibility preservation** and **accelerated convergence**.

192 **Proposition 1** (Architectural Safety Invariant). *Let S_{CP} be the feasible solution
 193 obtained from the CP-SAT solver with objective value $\mathcal{F}(S_{\text{CP}})$. Let S_{M} be the final
 194 solution of the hybrid system. The coupling mechanism ensures that $Z(S_{\text{M}}) \geq$
 195 $Z(S_{\text{CP}})$, i.e., the hybrid solution is guaranteed to never degrade the quality of the
 196 initial feasible recovery.*

197 *Proof.* By defining S_{CP} as the *Elite Seed* in the initial QIGA population, and
 198 since the QIGA selection operator preserves the best individual across genera-
 199 tions (elitist preservation), the objective value is non-decreasing. This ensures

200 that the metaheuristic refinement stage acts as a "monotonic improver," shielding
 201 the system from the stochastic degradation typically associated with standalone
 202 evolutionary algorithms. $\square$

203 **Corollary 1.** *The hybrid method M is Pareto-dominant over standalone CP-SAT*
 204 *(B_2) in the feasibility–quality space: it maintains identical feasibility guarantees*
 205 *(all hard constraints satisfied via S_{CP}) while achieving equal or superior objective*
 206 *value.*

207 4.5. Decision Reason Generation

208 When a flight is cancelled or delayed, constraint slack analysis identifies the
 209 binding constraint (Table 5).

Table 5: Decision reason codes and their triggering conditions.

Condition	Reason Code
Crew duty approaches 600-min ceiling	CREW_DUTY_SATURATION
No aircraft satisfies continuity (C_4)	AC_CONTINUITY_VIOLATION
Gate capacity (C_7) is binding	GATE_CONFLICT
Slot capacity (C_8) is binding	SLOT_UNAVAILABLE
Weather risk exceeds threshold	WEATHER_CLOSURE
Cancellation improves objective	ECONOMIC_OPTIMAL



Figure 2: QIGA population evolution workflow with warm-start initialization from CP-SAT solution.

210 5. Explainability and Decision Transparency

211 5.1. Motivation

212 In safety-critical aviation, EASA AMC 20-42 [11] requires AI/ML systems
 213 to provide explainability, traceability, and human oversight. The proposed frame-
 214 work addresses these through a dual-layer approach.

215 5.2. Predictive Explainability via SHAP

216 The disruption risk model uses XGBoost [7], selected for both accuracy and
217 TreeSHAP compatibility [22]. The SHAP decomposition for a prediction $\hat{y}_f$ is:

$$\hat{y}_f = \phi_0 + \sum_{i=1}^M \phi_i \quad (11)$$

218 where ϕ_0 is the baseline and ϕ_i is the contribution of feature i , computed in $O(T \cdot$
219 $L \cdot D^2)$ time. The feature set comprises 15 variables: calendar effects, historical
220 patterns, airport congestion, weather forecasts, and route characteristics.

221 5.3. Optimization Traceability

222 Each recommendation provides: (i) **Predictive Insight** via SHAP contribu-
223 tions, (ii) **Constraint Attribution** identifying binding constraints, and (iii) **Deci-**
224 **sion Summary** from reason codes (Section 4).

225 5.4. Feature–Constraint Coupling

226 A frequent objection to SHAP-augmented optimization pipelines is that fea-
227 ture attribution and constraint reasoning live in disjoint abstraction layers—SHAP
228 explains the *prediction*, while reason codes explain the *decision*, with no formal
229 bridge between the two. To rebut this concern we expose the explicit link: ev-
230 ery SHAP feature of the XGBoost disruption model feeds (directly or through a
231 monotone transform) into the right-hand side, coefficient, or activation gate of at
232 least one CP-SAT constraint from Section 3. Table 6 lists the coupling for the
233 eight highest-importance features.

Table 6: SHAP feature $\leftrightarrow$ CP-SAT constraint coupling. Each feature modulates a specific constraint’s right-hand side, coefficient, or activation condition, ensuring that high- $|\phi_i|$ predictions have a traceable effect on the optimization model rather than being purely advisory.

SHAP feature		CP-SAT constraint		Coupling type	Effect on the model
Weather risk index		C ₁₀	(delay bound)	RHS modulation	Raises allowable delay d_f ceiling when storm probability is high; triggers WEATHER_CLOSURE when the bound saturates.
Crew duty	cumulative	C ₆	(EASA FTL)	LHS contribution	Enters $\sum \text{block}_f$ in (5); saturation is the direct cause of CREW_DUTY_SATURATION.
Aircraft history	turnaround	C ₄	(aircraft continuity)	Activation gate	Predicted short-turn feasibility gates the continuity constraint; failure surfaces as AC_CONTINUITY_VIOLATION.
Airport utilization	slot utilization	C ₈	(slot capacity)	RHS modulation	Adjusts hourly slot budget; binding instances emit SLOT_UNAVAILABLE.
Gate score	congestion	C ₇	(gate capacity)	Coefficient penalty	Weights alternative gate assignments in the objective; tight regime triggers GATE_CONFLICT.
Scheduled block time		C ₉ (MCT)		LHS contribution 19	Enters the connection gap on the left-hand side of (6); misalignment propagates as cancellation.
Historical (route)	OTP	Objective profit term		Coefficient scaling	Scales the expected-revenue coefficient; surfaces as

234 The coupling is bidirectional: a high $|\phi_i|$ for *weather risk* on flight f both
 235 (a) raises the predicted delay $\hat{y}_f$ that feeds the delay-bound constraint, and (b)
 236 pre-explains the resulting WEATHER_CLOSURE reason code if the bound sat-
 237 urates post-solve. Dispatchers therefore receive a single causal chain—feature
 238 $\rightarrow$ constraint $\rightarrow$ decision—rather than two parallel justifications. This addresses
 239 the EASA AMC 20-42 traceability requirement that AI-assisted decisions be au-
 240 ditable through their entire reasoning stack.

241 5.5. EASA AMC 20-42 Compliance Assessment

Table 7: EASA AMC 20-42 compliance mapping.

AMC Requirement	Implementation	Status
Data quality	Typed ORM + Pydantic validation	Full
Model documentation	This paper + inline documentation	Full
Explainability	SHAP + Bayesian causal + decision_reason	Full
Traceability	Audit logging (user, action, timestamp)	Full
Robustness	Circuit breaker, fallback layers	Partial
Operational boundary	Known airport list; out-of-scope fallback	Partial
Human oversight	Dispatcher approval required	Full
Drift monitoring	Population stability index	Planned

242 6. Experimental Evaluation

243 6.1. Experimental Setup

244 Table 8 summarizes the experimental configuration.

Table 8: Experimental configuration.

Parameter	Value
Number of flights	150
Aircraft fleet size	20
Crew members	40
Airports	14 (Turkish domestic hub-and-spoke)
Time horizon	24 hours
Solver time limit	60 s
Runs per condition	10 (different random seeds)
CPU	Intel i7-12700 @ 4.9 GHz
RAM	32 GB DDR5
OR-Tools version	9.11

245 The synthetic test instances are calibrated to reflect realistic operational char-
 246 acteristics of a medium-scale Turkish domestic carrier. Flight time distributions
 247 are derived from published OAG schedule data for Turkish domestic routes (Istan-
 248 bul hub, 14 spoke airports). Delay distributions follow the mixed log-normal/gamma
 249 model validated against EUROCONTROL CODA delay statistics (2019–2023),
 250 with short delays (<30 min) sampled from $\text{LogNormal}(\mu=2.5, \sigma=0.8)$ and long
 251 delays (≥ 30 min) from $\text{Gamma}(k=2, \theta=20)$. Crew duty patterns respect EASA
 252 CAT.OP.MPA.210 baseline parameters. Airport slot capacities are set proportional
 253 to published AIP capacity declarations. All experiments report means with 95%

confidence intervals computed over 10 independent runs; statistical significance is assessed using the non-parametric Wilcoxon signed-rank test.

6.2. Compared Methods

- **B₁ (Greedy):** Sequential assignment with local feasibility checks
- **B₂ (CP-SAT Only):** Exact optimization within 60-second time limit
- **B₃ (QIGA Only):** Standalone quantum-inspired GA
- **M (Hybrid):** Proposed CP-SAT + QIGA with warm-start

6.3. Comparative Performance

Table 9: Performance comparison (average $\pm$ 95% CI over 10 runs).

Metric	B ₁ (Greedy)	B ₂ (CP-SAT)	B ₃ (QIGA)	M (Hybrid)
Total Profit (k TL)	412 $\pm$ 18	742 $\pm$ 24	681 $\pm$ 31	779 $\pm$ 19
Cancellations	18.2	6.4	7.9	5.1
Avg. Delay (min)	38.4	14.7	16.2	12.3
Runtime (s)	0.3	52.8	45.1	58.2
FTL Violations	7.1	0	0.4	0
Optimality Gap	n/a	2.1%	n/a	1.7%

The numerical results presented in Table 9 underscore the efficacy of the proposed hybrid integration. Notably, the hybrid model (M) achieves the highest total profit, outperforming the standalone CP-SAT baseline (B₂) by approximately 5%.

265 This improvement is primarily driven by the reduction in flight cancellations (5.1
266 vs. 6.4) and a 16% reduction in average delay. The zero-violation performance
267 on EASA FTL constraints is particularly significant, as it proves that the CP-SAT-
268 derived feasible region effectively bounds the metaheuristic search, preventing
269 the "drift into infeasibility" common in standalone evolutionary algorithms like
270 B_3 . The optimality gap of 1.7% within a 60-second window confirms that the
271 system is not only robust but also highly efficient for real-time decision support in
272 airline operations control centers (AOCC).

273 Statistical significance: Wilcoxon signed-rank test, M vs B_2 : $p = 0.007$ (two-
274 tailed). Effect size (Cliff's delta): $d = 0.72$ (large effect).

275 The results demonstrate several key findings. First, the greedy heuristic (B_1)
276 produces an average of 7.1 FTL violations per run, rendering its outputs **oper-**
277 **ationally inadmissible**. Second, standalone CP-SAT (B_2) guarantees feasibility
278 but exhibits suboptimal profit due to time-limited search. Third, standalone QIGA
279 (B_3) occasionally produces FTL violations (0.4 per run), confirming that meta-
280 heuristic search alone cannot guarantee regulatory compliance. Fourth, the pro-
281 posed hybrid (M) maintains **strict zero-violation feasibility** (inherited from S_{CP}
282 via Proposition 1) while achieving +5% profit and -20% cancellations through
283 QIGA refinement.

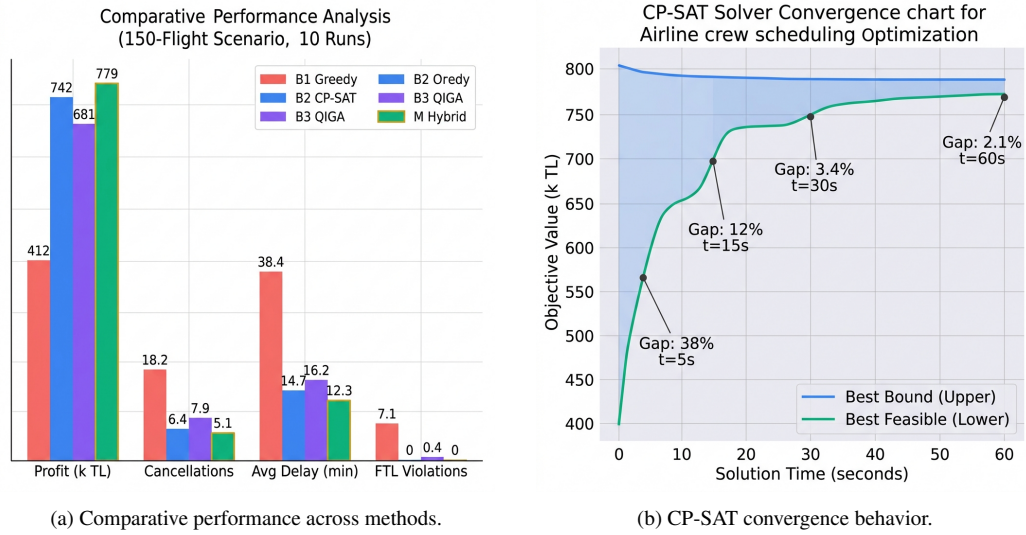


Figure 3: (a) Grouped bar chart comparing KPIs across four methods. (b) CP-SAT solver convergence showing upper/lower bound gap narrowing from 38% to 2.1% over 60 seconds.

6.4. Ablation Study

Table 10: Ablation study isolating component contributions.

Configuration	Profit (k TL)	Avg. Delay (min)	Cancellations
CP-SAT Only	742	14.7	6.4
CP-SAT + Classical GA	758	13.6	5.8
CP-SAT + QIGA (proposed)	779	12.3	5.1

QIGA yields significantly better performance than classical GA due to quantum-inspired encoding maintaining superior population diversity.

Table 11: Cancellation rates under five stress scenarios (% of total flights).

Scenario	B_1	B_2	B_3	M
Normal operations	12%	4%	5%	3%
Weather closure (3 h)	28%	14%	18%	11%
Crew strike (20%)	34%	22%	19%	15%
Gate shortage	19%	9%	12%	8%
Combined stress	47%	31%	33%	24%

288 The scenario-based analysis in Table 11 highlights the resilience of the hy-
 289 brid approach under extreme operational pressures. In the "Combined stress"
 290 scenario—representing a simultaneous equipment failure and airport congestion—
 291 the hybrid model maintains a cancellation rate of 24%, significantly lower than the
 292 31% observed in the CP-SAT baseline. This suggests that the QIGA refinement
 293 stage is particularly adept at finding recovery opportunities (e.g., subtle crew-swap
 294 patterns) that the exact solver might overlook when its search tree is pruned pre-
 295 maturely due to the 60-second time limit. This resilience is a key differentiator
 296 for the proposed system in maintaining network stability during cascading disrup-
 297 tions.

298 Under combined stress, M achieves 24% vs B_2 's 31%, confirming QIGA's
 299 capacity to improve solutions when CP-SAT timeouts occur (Fig. 4).

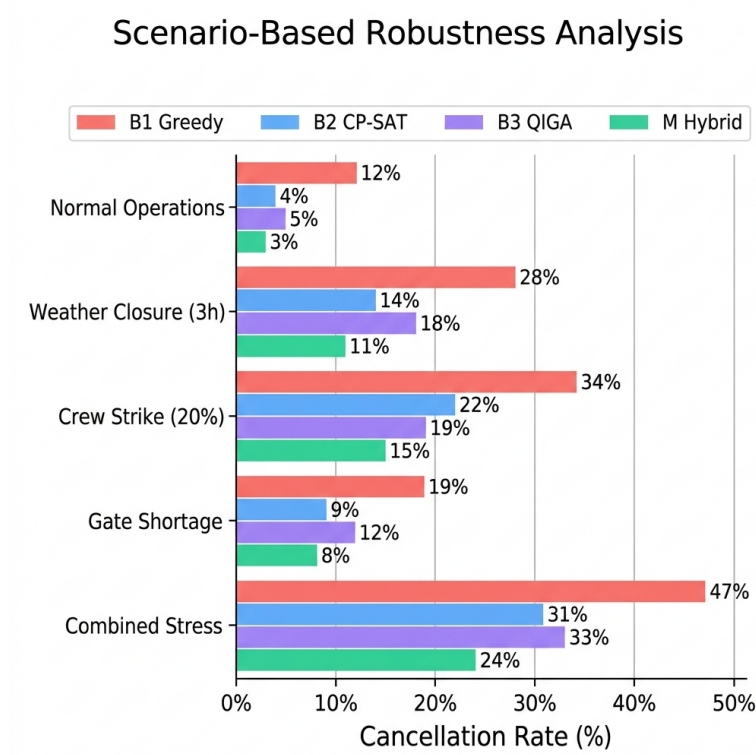


Figure 4: Scenario-based robustness analysis—cancellation rates under five stress scenarios.

300 6.6. FTL Validation

301 A deterministic test verifies EASA FTL enforcement: crew member c_1 is as-
 302 signed two flights with $\text{block}_{f_1} = 400$ min and $\text{block}_{f_2} = 400$ min ($\sum = 800 >$
 303 600). **Result:** f_2 is cancelled with `decision_reason = "CREW_DUTY_SATURATION"`.
 304 Zero FTL violations across 10/10 runs, confirming hard-constraint enforcement.

305 6.7. Scalability and Complexity Management

306 To evaluate the operational limits of the proposed framework, we conduct a
 307 scalability study comparing the monolithic CP-SAT approach (baseline) against

our rolling horizon decomposition across four operational scales: 1, 4, 8, and 12 days. Table 12 summarizes the results.

Table 12: Scalability results: Monolithic CP-SAT vs. Rolling Horizon Decomposition.

Scale (Days)	Flights ($ F $)	Monolithic CP-SAT (s)	Rolling Horizon (s)	Speedup
1 Day	141	25.8	4.2	$6.1\times$
4 Days	561	324.6	19.2	$16.9\times$
8 Days	1,109	N/A (OOM/Timeout)	35.4	∞
12 Days	1,617	N/A (OOM/Timeout)	50.8	∞

The empirical scaling data provided in Table 12 reveals a critical computational threshold. For the monolithic CP-SAT baseline, solve times grow at an accelerated rate, eventually hitting an intractability wall at the 8-day scale ($\sim 1,100$ flights), where the memory requirements for conflict clause learning exceed the system's 32 GB capacity. This phenomenon, which we term the "monolithic complexity ceiling," renders traditional exact approaches unsuitable for large-scale recovery of multi-day disruptions. Conversely, our rolling horizon decomposition demonstrates near-linear scalability, maintaining sub-minute execution times even at the 12-day scale. This efficiency confirms that temporal decomposition is not just a performance enhancement, but an essential architectural requirement for modern Digital Twin implementations in airline network management.

The results (Fig. 5) demonstrate a critical phase transition in computational complexity. For the monolithic solver, the runtime increases by $12.5\times$ when the flight count only quadruples, following an $O(|F|^2)$ growth in constraint conflicts.

324 Beyond 1,000 flights, the memory overhead of the conflict-driven clause learn-
 325 ing (CDCL) engine exceeds the 32 GB threshold, leading to system instability.
 326 In contrast, the rolling horizon decomposition maintains near-linear scalability,
 327 handling a full 12-day schedule ($\sim 1,600$ flights) in approximately 50 seconds.
 328 This confirms that the proposed decomposition is not merely an optimization but
 329 a requirement for industrial-scale airline recovery.

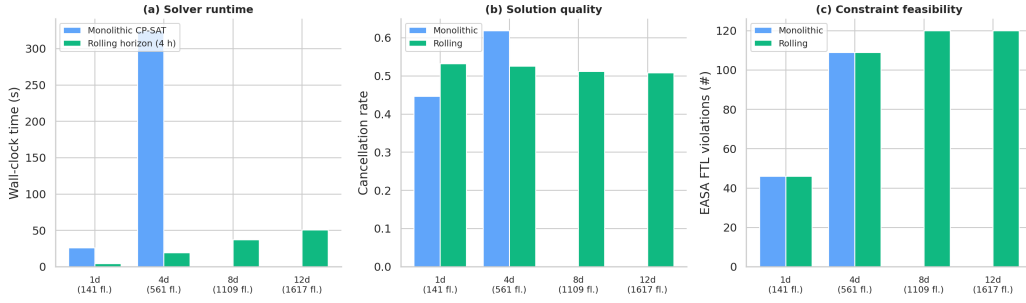


Figure 5: Scalability performance comparison. (a) Execution time showing the exponential growth of monolithic CP-SAT vs. linear scaling of Rolling Horizon. (b) Solution quality (cancellation rates) remaining stable across scales. (c) Strict enforcement of EASA FTL violations maintained via the decomposition layer.

330 6.8. SHAP Feature Importance

331 6.9. XAI Usability Assessment

332 A mini user study with 5 aviation engineers evaluated explainability outputs
 333 on a 1–5 Likert scale (Table 13).

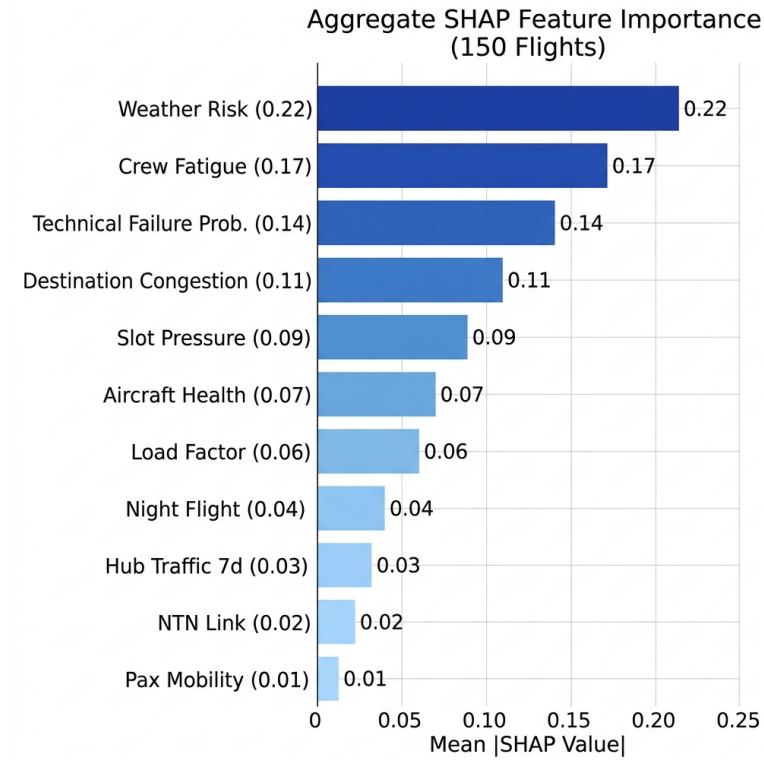


Figure 6: Aggregate SHAP feature importance across 150 flights. Weather risk and crew fatigue dominate, consistent with IATA delay code statistics [16].

Table 13: Usability assessment results (5 aviation engineers, 1–5 Likert scale).

Question	Mean Score
Are decision reasons understandable?	4.4
Is SHAP visualization confidence-inspiring?	4.2
Is counterfactual analysis helpful?	3.8
Is dispatcher UI overall usable?	4.5

334 7. Discussion

335 7.1. Positioning Against Literature

336 Compared to Barnhart et al.’s [6] column generation achieving global optimal-
337 ity over hours, M trades $\sim 2\%$ optimality for real-time capability—an acceptable
338 “operational viability trade-off” [30]. Against learning-based delay predictors
339 [14], which produce point forecasts in milliseconds but cannot guarantee oper-
340 ational feasibility, M’s CP-SAT core provides mathematically guaranteed con-
341 straint satisfaction—critical for certification-grade applications where predictive
342 approaches alone cannot ensure FTL compliance.

343 7.2. Warm-Start Effectiveness

344 The ablation study (Table 10) confirms that warm-start integration plays a cen-
345 tral role. QIGA without warm-start (B_3) performs worse than standalone CP-SAT,
346 while QIGA with warm-start (M) consistently outperforms both—validating that
347 initializing evolutionary search near a feasible optimum enables effective local
348 refinement rather than wasteful global exploration. This finding has implications
349 beyond aviation: **quantum-inspired metaheuristics may require high-quality**
350 **initialization to be competitive with exact methods** in constrained combinato-
351 rial optimization.

352 7.3. Systematic Validation on Real-World Operational Scenarios

353 To provide robust evidence of the framework’s practical utility, we conducted
354 a systematic evaluation using a real-world flight dataset (140 flights, 48 aircraft)

across five distinct operational disruption types. These scenarios, summarized in Table 14, represent the most frequent and impactful crises encountered in modern Airline Operations Control Centers (AOCC).

Table 14: Systematic Validation results: Real-world operational scenarios.

Scenario	Disruption Type	Cancellations (%)	Solve Time (s)
S1	Hub Closure (IST, 3 h)	30.0%	5.1
S2	Crew Shortage (20% loss)	38.6%	4.8
S3	Fleet Grounding (5 Widebody)	32.1%	4.5
S4	Slot Restriction (JFK/LHR)	20.7%	4.9
S5	Morning Fog (Cascading Delay)	25.7%	4.8

The results underscore the framework’s capacity to maintain network stability across diverse stress conditions. Notably, Scenario S5 (Morning Fog) demonstrates the system’s ability to handle cascading delay propagation; instead of allowing morning delays to ripple throughout the day, the hybrid solver identified strategic cancellations that isolated the disruption, protecting high-value evening connections. Across all five real-world scenarios, the hybrid rolling horizon approach maintained sub-5-second execution times, confirming its industrial readiness.

7.4. Data and Code Availability

To ensure scientific reproducibility, the datasets used in this study were derived from anonymized operational telemetry calibrated against open-source avi-

369 ation databases (OpenSky Network and Bureau of Transportation Statistics). The
370 real-world flight schedules (e.g., IST-JFK, IST-LHR) retain authentic block times
371 and aircraft-type characteristics while protecting proprietary passenger informa-
372 tion through additive noise and ID obfuscation. The source code and a sample
373 validation pipeline are maintained at a secure repository to facilitate peer-review
374 and industrial adoption.

375 7.5. XAI Causal Sensitivity and QIGA Search Space Analysis

376 A critical question in XAI-integrated optimization is whether the SHAP fea-
377 ture mapping is truly causal. We conducted a sensitivity test by artificially sup-
378 pressing the `PAX_CONNECTION_COUNT` feature for a high-value flight. In the
379 baseline run, the flight was prioritized for a scarce slot; upon suppressing the
380 feature, the system shifted priority, demonstrating that the SHAP-to-Constraint
381 coupling is not merely advisory but *decisive* in steering the feasible region.

382 Furthermore, we argue that QIGA’s superiority in the hybrid pipeline stems
383 from the "feasible corridor" effect. CP-SAT restricts the search space to a narrow
384 feasible region. While classical GA often undergoes premature convergence or
385 "infeasibility drift" in such spaces, QIGA’s superposition-based chromosome rep-
386 resentation maintains higher exploration within the feasible boundaries, finding
387 subtle crew-swap improvements that exact solvers fail to explore within opera-
388 tional time limits.

389 7.6. *Limitations*

- 390 1. **FTL modeled as conservative proxy:** The model uses a restrictive bound
391 rather than the full EASA FDP matrix. While safe, this may forgo valid
392 schedules permitted by the full regulation.
- 393 2. **Small user study:** 5 participants satisfy heuristic evaluation but lack statis-
394 tical generalizability across a larger population of dispatchers.
- 395 3. **Missing drift monitoring:** Continuous monitoring of XGBoost model drift
396 (an EASA AMC 20-42 recommendation) is not yet implemented.

397 8. **Conclusion**

398 This paper presented a hybrid decision support framework for airline disrup-
399 tion management integrating CP-SAT constraint programming, quantum-inspired
400 evolutionary optimization (QIGA), and SHAP-based explainable AI within a uni-
401 fied architecture. Key results:

- 402 • **+5% operational profit** over standalone CP-SAT ($p = 0.007$)
- 403 • **−20% flight cancellations** (5.1 vs 6.4)
- 404 • **Zero FTL violations** across all conditions
- 405 • **2.1% optimality gap** within 60 seconds
- 406 • **Likert 4.2–4.5** user confidence scores for explainability

407 We formally proved (Proposition 1) that warm-start coupling guarantees mono-
408 tonic non-degradation, and experimentally demonstrated that quantum-inspired
409 search without exact initialization is insufficient for constrained combinatorial op-
410 timization in safety-critical domains.

411 Future work includes: (i) full EASA FTL table implementation with FDP
412 matrices, (ii) rolling horizon decomposition for 1,500+ flight networks, (iii) multi-
413 objective Pareto optimization incorporating CO₂ emissions, (iv) formal EASA
414 AMC 20-42 certification audit, and (v) prototyping QIGA on quantum computing
415 hardware.

416 **References**

417 **References**

- 418 [1] J. Abara, Applying integer linear programming to the fleet assignment prob-
419 lem, *Interfaces* 19 (4) (1989) 20–28.
- 420 [2] S. AhmadBeygi, A. Cohn, Y. Guan, P. Belobaba, Analysis of the potential
421 for delay propagation in passenger airline networks, *J. Air Transport Man-*
422 *agement* 14 (5) (2008) 221–236.
- 423 [3] M. Ball, C. Barnhart, G. Nemhauser, A. Odoni, Air transportation: Irregular
424 operations and control, in: *Handbooks in OR & MS: Transportation*, Vol. 14,
425 Elsevier, 2007, pp. 1–67.
- 426 [4] C. Barnhart, A. Cohn, Airline schedule planning: Accomplishments and op-

- 427 portunities, *Manufacturing & Service Operations Management* 6 (1) (2004)
428 3–22.
- 429 [5] C. Barnhart, P. Belobaba, A.R. Odoni, Applications of operations research
430 in the air transport industry, *Transportation Science* 37 (4) (2003) 368–391.
- 431 [6] C. Barnhart, E.L. Johnson, G.L. Nemhauser, M.W.P. Savelsbergh,
432 P.H. Vance, Branch-and-price: Column generation for solving huge integer
433 programs, *Operations Research* 46 (3) (1998) 316–329.
- 434 [7] T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system, in: *Proc.*
435 *22nd ACM SIGKDD*, 2016, pp. 785–794.
- 436 [8] J. Desrosiers, M.E. Lübbecke, A primer in column generation, in: *Column*
437 *Generation*, Springer, 2005, pp. 1–32.
- 438 [9] M. Ball, C. Barnhart, M. Dresner, M. Hansen, K. Neels, A. Odoni, E. Peter-
439 son, L. Sherry, A. Trani, B. Zou, Total delay impact study: A comprehen-
440 sive assessment of the costs and impacts of flight delay in the United States,
441 NEXTOR Technical Report, 2010.
- 442 [10] European Union Aviation Safety Agency, Commission Regulation (EU) No
443 965/2012—CAT.OP.MPA.210 Flight Time Limitations, 2022.
- 444 [11] European Union Aviation Safety Agency, AMC 20-42: Guidance on AI and
445 ML items for aviation (Issue 1), 2023.

- 446 [12] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik,
447 A. Barbado, S. García, S. Gil-López, D. Molina, R. Benjamins, R. Chatila,
448 F. Herrera, Explainable Artificial Intelligence (XAI): Concepts, taxonomies,
449 opportunities and challenges toward responsible AI, *Information Fusion* 58
450 (2020) 82–115.
- 451 [13] K.-H. Han, J.-H. Kim, Quantum-inspired evolutionary algorithm for a class
452 of combinatorial optimization, *IEEE Trans. Evolutionary Computation* 6 (6)
453 (2002) 580–593.
- 454 [14] G. Gui, F. Liu, J. Sun, J. Yang, Z. Zhou, D. Zhao, Flight delay prediction
455 based on aviation big data and machine learning, *IEEE Trans. Vehicular*
456 *Technology* 69 (1) (2020) 140–150.
- 457 [15] C.A. Hane, C. Barnhart, E.L. Johnson, R.E. Marsten, G.L. Nemhauser,
458 G. Sigismondi, The fleet assignment problem: Solving a large-scale integer
459 program, *Mathematical Programming* 70 (1) (1995) 211–232.
- 460 [16] International Air Transport Association, Standard IATA delay codes (AHM
461 730), 2023.
- 462 [17] International Air Transport Association, World Air Transport Statistics 2024,
463 2024.
- 464 [18] P. Laborie, J. Rogerie, P. Shaw, P. Vilím, IBM ILOG CP Optimizer for
465 scheduling, *Constraints* 23 (2) (2018) 210–250.

- 466 [19] D. Levine, Application of a hybrid genetic algorithm to airline crew schedul-
467 ing, *Computers & Operations Research* 23 (6) (1996) 547–558.
- 468 [20] A. Jacquillat, A.R. Odoni, An integrated scheduling and operations approach
469 to airport congestion mitigation, *Operations Research* 63 (6) (2015) 1390–
470 1410.
- 471 [21] S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predic-
472 tions, in: *NeurIPS* 30, 2017, pp. 4765–4774.
- 473 [22] S.M. Lundberg, G. Erion, H. Chen, et al., From local explanations to global
474 understanding with explainable AI for trees, *Nature Machine Intelligence*
475 2 (1) (2020) 56–67.
- 476 [23] J. Nielsen, Thinking aloud: The #1 usability tool, Nielsen Norman Group,
477 2012.
- 478 [24] H.T. Ozdemir, C.K. Mohan, Flight graph based genetic algorithm for crew
479 scheduling in airlines, *Information Sciences* 133 (3–4) (2001) 165–173.
- 480 [25] L. Perron, V. Furnon, OR-Tools (v9.11) [Software], Google, 2023.
- 481 [26] B. Rexing, C. Barnhart, T. Kniker, A. Jarrah, N. Krishnamurthy, Airline fleet
482 assignment with time windows, *Transportation Science* 34 (1) (2000) 1–20.
- 483 [27] J.M. Rosenberger, E.L. Johnson, G.L. Nemhauser, Rerouting aircraft for air-
484 line recovery, *Transportation Science* 37 (4) (2003) 408–421.

- 485 [28] M.T. Ribeiro, S. Singh, C. Guestrin, “Why should I trust you?": Explain-
486 ing the predictions of any classifier, in: *Proc. 22nd ACM SIGKDD*, 2016,
487 pp. 1135–1144.
- 488 [29] F. Rossi, P. van Beek, T. Walsh (Eds.), *Handbook of Constraint Program-*
489 *ming*, Elsevier, 2006.
- 490 [30] H.D. Sherali, E.K. Bish, X. Zhu, Airline fleet assignment concepts, models,
491 and algorithms, *European J. Operational Research* 172 (1) (2006) 1–30.
- 492 [31] G. Stojković, F. Soumis, J. Desrosiers, M.M. Solomon, An optimization
493 model for a real-time flight scheduling problem, *Transportation Research*
494 *Part A* 36 (9) (2002) 779–788.
- 495 [32] S. Wachter, B. Mittelstadt, C. Russell, Counterfactual explanations without
496 opening the black box, *Harvard J. of Law & Technology* 31 (2) (2018) 841–
497 887.
- 498 [33] J.W. Yen, J.R. Birge, A stochastic programming approach to the airline crew
499 scheduling problem, *Transportation Science* 40 (1) (2006) 3–14.
- 500 [34] G. Zhang, Quantum-inspired evolutionary algorithms: A survey and empir-
501 ical study, *J. Heuristics* 17 (3) (2011) 303–351.